

MATH 162: Calculus

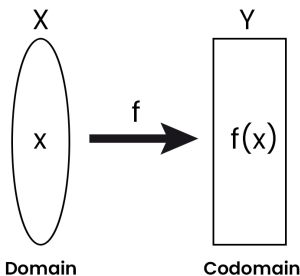
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Functions

A function is a rule that assigns to an element of a set X , one and only one element in a set Y . The set X is called the domain of the function and Y is called the co-domain of the function. Symbolically, a function f is written as $f : X \longrightarrow Y$.

For each element $x \in X$, $f(x) \in Y$ is the image of x



There are instances where some element in the co-domain, Y may not serve as images for elements in the domain X . If every element in Y is an image for some element in X then Y is called the range of the function.

The range of a function is the set of all images. The range of a function is a subset of the co-domain.

A function $f : A \longrightarrow B$ can also be defined as a set of ordered pairs; $\{(a, b) : a \in A \text{ and } b \in B\}$



Example

Let $f : A \longrightarrow B$ be defined by $f(x) = 2x + 3$. Evaluate the following

1. $f(0)$
2. $f(1)$
3. $f(-3)$
4. $f(a)$

Example

A function is defined by $f : x \longrightarrow \frac{3x-1}{x-3}$ on $\{-2, -1, 0, 1, 2, \}$.

1. Find the range of f
2. What values of x make the function undefined?



Determination of the Domain of a Function

To determine the domain of a function, we exclude all possible values of x which makes the function undefined. i.e.

1. The values of x which make the denominator of a rational function zero.
2. The values of x which produce a negative sign under an even indexed radical



Example

State the largest possible domain of the following functions.

1. $f(x) = \frac{1-2x}{\sqrt{2x-1}}$

2. $f : x \longrightarrow \frac{x+5}{x-3}$

3. $f(x) = 1 - 2x$

4. $g : x \longrightarrow \sqrt{2x-1}$

5. $g : x \longrightarrow \frac{x}{\sqrt{2x-1}}$

6. $g(x) = \sqrt{1-x^2}$

7. $f(x) = \sqrt{5x-2}$

8. $f(x) = \frac{\sqrt{x-2}}{x^2-4}$



Limit

The two main branches of calculus; differential calculus and integral calculus, depend on the limit concept.

To help grasp the idea on which calculus is based, we first provide a practical introduction of limit of a function.



The Meaning of $x \longrightarrow a$

For a given function f , $x \longrightarrow a$ gives the behaviour of $f(x)$ for values of x around a . Consider the behaviour of $f(x) = 2x + 3$ around 2.

x	1	1.9	1.99	1.999	1.9999	2	2.0001	2.001	2.01	2.1	3
f(x)	5	6.8	6.98	6.998	6.9998		7.0002	7.002	7.02	7.2	9

From the table above, we see clearly that $f(x)$ starts at 5 and increases progressively to 6.9998 as we move from left to right approaching 2. Again, $f(x)$ starts from 9 and reduces progressively to 7.0002 as we move from right to left approaching 2.



Therefore as we allow x to approach 2, the values of $f(x)$ approach 7.

We say that "as x approaches 2, $f(x)$ approaches 7" or "as x tends to 2, $f(x)$ tends to 7."

The mathematical notation for this is expressed as

$$\lim_{x \rightarrow 2} (2x + 3) = 7 \text{ or } 2x + 3 \rightarrow 7 \text{ as } x \rightarrow 2.$$

The number 7 is called the limit of $f(x)$ as $x \rightarrow 2$.



Examples

Evaluate each of the following limits.

$$1. \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

$$2. \lim_{x \rightarrow 4} (2x + 3)$$

$$3. \lim_{x \rightarrow 1} \left\{ \frac{x^2 - 1}{x + 1} \right\}$$

$$4. \lim_{x \rightarrow 2} \left\{ \frac{x^2 - 4}{x - 2} \right\}$$

$$5. \lim_{x \rightarrow 4} \left\{ \frac{x^2 - 7x + 10}{x^2 - 4} \right\}$$

$$6. \lim_{x \rightarrow 1} \left(\frac{x^3 - 1}{x - 1} \right)$$

$$7. \lim_{x \rightarrow 9} \left\{ \frac{9 - x}{3 - \sqrt{x}} \right\}$$

$$8. \lim_{x \rightarrow 3} \left\{ \frac{9 - x^2}{4 - \sqrt{x^2 + 7}} \right\}$$



Left-hand and Right-hand Limits

$\lim_{x \rightarrow a^-} f(x)$ indicates "the limit of $f(x)$ as x approaches a from the left of a ". This is called left-hand limit.

Also, $\lim_{x \rightarrow a^+} f(x)$ indicates " the limit of $f(x)$ as x approaches a from the right of a . This is called the right-hand limit.

Definition

For any function f , $\lim_{x \rightarrow a} f(x) = L$ if and only if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$



Remark

1. If the left-hand limit does not equal the right-hand limit, then there is no limit. Under such circumstances, we say that the limit does not exist.
2. The existence of $\lim_{x \rightarrow a} f(x)$ does not depend on whether $f(a)$ is defined.
3. The existence of $\lim_{x \rightarrow a} f(x)$ does not depend on the value $f(a)$ if $f(a)$ is defined.



Limit Theorems

If a, c and n are real numbers then

$$1. \lim_{x \rightarrow a} c = c$$

$$2. \lim_{x \rightarrow a} x = a$$

$$3. \lim_{x \rightarrow a} c f(x) = c \lim_{x \rightarrow a} f(x)$$

$$4. \lim_{x \rightarrow a} \{f(x) + g(x)\} = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$5. \lim_{x \rightarrow a} \{f(x) - g(x)\} = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$6. \lim_{x \rightarrow a} \{f(x) \cdot g(x)\} = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$7. \lim_{x \rightarrow a} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} ; \lim_{x \rightarrow a} \{g(x)\} \neq 0$$

$$8. \lim_{x \rightarrow a} \{f(x)\}^n = \left\{ \lim_{x \rightarrow a} f(x) \right\}^n$$



Examples

Compute each of the following limits.

$$1. \lim_{x \rightarrow 2} (x + 3)(x^2 - 5)$$

$$2. \lim_{x \rightarrow 2} f(x) \text{ when } f(x) = \begin{cases} 1 - x, & x \leq 2 \\ 4, & x > 2 \end{cases}$$

$$3. \lim_{x \rightarrow 0} \frac{|x|}{x}$$

$$4. \lim_{x \rightarrow 1} f(x) \text{ when } f(x) = \begin{cases} x + 1, & x \neq 1 \\ \pi, & x = 1 \end{cases}$$



5. Determine the existence or otherwise of $\lim_{x \rightarrow 4} f(x)$ when

$$f(x) = \begin{cases} \sqrt{x-4}, & \text{if } x > 4 \\ 8 - 2x, & \text{if } x < 4 \end{cases}$$

6. Evaluate $\lim_{x \rightarrow 1} f(x)$ when $f(x) = \begin{cases} \frac{x-1}{x^2-1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$

7 Show that $\lim_{x \rightarrow 0} |x| = 0$.



Limit of the Form $\lim_{x \rightarrow 0} \frac{1}{x^n}$

As x gets closer to 0, x^n also gets closer to 0, and $\frac{1}{x^n}$ becomes very large. Consider $f(x) = \frac{1}{x^2}$ as $x \rightarrow 0$.

x	-1	-0.5	-0.2	-0.1	-0.05	-0.01	-0.001	0	0.001	0.01	0.05	0.1	0.2	0.5	1
f(x)	1	4	25	100	400	10000	1000000		1000000	10000	400	100	25	4	1

As clearly shown above, $f(x)$ can be made arbitrarily large by taking x close enough to 0. Then the values of $f(x)$ do not approach a number, so $\lim_{x \rightarrow 0} \frac{1}{x^n}$ does not exist. We use the notation $\lim_{x \rightarrow 0} \frac{1}{x^n} = \infty$ to describe the behaviour of $f(x)$ as $x \rightarrow 0$.



Remark

$\lim_{x \rightarrow 0} \frac{1}{x^n} = \infty$ does not mean we are regarding ∞ as a number, or the limit exists. It simply expresses the particular way in which the limit does not exist. $f(x) = \frac{1}{x^2}$ can be made as large as we like by taking x close to zero.

In general, we write symbolically $\lim_{x \rightarrow a} f(x) = \infty$ to indicate that the values of $f(x)$ become larger and larger (or increases without bound) as x becomes closer and closer to a .



Limits of the Form $\lim_{x \rightarrow \infty} f(x)$

The limit of a function $f(x)$ as x approaches infinity (∞) or becomes very large is denoted as $\lim_{x \rightarrow \infty} f(x)$.

1. $\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$ if the index, n is positive.
2. $\lim_{x \rightarrow \infty} x^n$ does not exist if the index n is positive.
3. $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ can be found by first dividing the numerator and the denominator by the variable with the highest index of x .



Example

Evaluate the following limits.

$$1. \lim_{x \rightarrow \infty} \left(\frac{5x^2 - 1}{2x^2 + 1} \right)$$

$$2. \lim_{x \rightarrow \infty} \left(\frac{7x}{x - 4} \right)$$

$$3. \lim_{x \rightarrow \infty} \left(\frac{2x^3 - 7x + 3}{x^3 + 2} \right)$$

$$4. \lim_{x \rightarrow \infty} \left(\frac{3x}{x + 2} \right)$$



Limit of Simple Trigonometric Functions

Consider the function $f(x) = \frac{\sin x}{x}$, where x is in radian.

x	2	1.5	1.0	0.5	0.3	0.2	0.1	0.01	0.001	0
f(x)	0.45	0.66	0.84	0.96	0.99	0.99	0.998	0.999	0.999	?

As $x \longrightarrow 0$, $\frac{\sin x}{x} \longrightarrow 1$, thus $\lim_{x \longrightarrow 0} \frac{\sin x}{x} = 1$.

We note the following.

1. $\lim_{x \longrightarrow 0} \cos x = 1$
2. $\lim_{x \longrightarrow 0} \sin x = 0$
3. $\lim_{x \longrightarrow 0} \tan x = 0$
4. $\lim_{x \longrightarrow 0} \frac{\tan x}{x} = 1$
5. $\lim_{x \longrightarrow 0} \frac{\cos x}{x} = 0$
6. $\lim_{x \longrightarrow 0} \frac{\sin x}{x} = 1$



Examples

1. Show that $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
2. Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$
3. Compute $\lim_{x \rightarrow 0} \sec x$

Example

Evaluate the following limits.

1. $\lim_{x \rightarrow 0} (\cos x + \sin x)$
2. $\lim_{x \rightarrow 0} \left(\frac{2 \tan x - x}{\sin 2x} \right)$
3. $\lim_{x \rightarrow 0} \left(\frac{\sqrt{2} - \sin x}{\cos x} \right)$
4. $\lim_{x \rightarrow 0} \left(\frac{\sin 5x}{x} \right)$
5. $\lim_{x \rightarrow 0} \left(\frac{\cot x}{\cot 2x} \right)$



Calculus

Calculus is the mathematical study of change. It has two major branches.

1. Differential calculus, which deals with the rate of change and slope of curves.
2. Integral calculus, which is concerned with the accumulation of quantities and the areas under curves.



The Fundamental Theorem of Calculus

This theorem states that if a function f is continuous on an interval (a, b) , and if F is a function whose derivative is f on the interval (a, b) , then $\int_a^b f(x) dx = F(b) - F(a)$.

Furthermore, for any x in the interval (a, b) ,

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

.

The fundamental theorem of calculus simply says that differentiation and integration are inverse/reverse operations.



Continuity of Real-Valued Functions

A function f is said to be continuous if its graph has no breaks in it such as holes, gaps, or jumps. This means the graph of the function can be drawn without lifting the pencil off the paper. If the graph of the function has a hole, a gap, or a jump at $x = a$, we say that the function is discontinuous at $x = a$.

Definition

A function f is said to be continuous at a point $x = a$ if the following conditions are satisfied.

1. $f(a)$ is defined or f is defined at a .
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $f(a) = \lim_{x \rightarrow a} f(x)$.



Definition

A function $f(x)$ is continuous on an interval (a, b) if it is continuous for all values of x in the interval (a, b) .

Examples

1. Show that $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$
2. verify the continuity of $f(x) = \frac{x^2 - 4}{x - 2}$ at $x = 2$.



Remark

1. Every polynomial function is continuous for all real x .
2. A rational function is continuous for all real x except those values of x for which the denominator equals zero.
3. Every exponential function is continuous for all real x .
4. The natural logarithmic function, $\ln |x|$ is continuous for all real x in its domain.
5. A radical function, or rational exponent function is continuous for all real x in its domain.



Example

Determine the continuity of the given function at the point indicated.

1. $f(x) = 3x - 2$; at $x = -2$

2. $f(x) = 2x + 6$; at $x = -3$

3. $h(x) = x^2 - x - 6$; at $x = 3$

4. $h(x) = 2x^3x^2 - 5$; at $x = 2$

5. $g(x) = \frac{x+1}{x-3}$; at $x = 3$

6. $g(x) = \frac{x-5}{x+2}$; at $x = -2$

7. $f(x) = \frac{x^2-25}{x-5}$; at $x = 5$

8. $f(x) = \frac{x^2-25}{x-5}$; at $x = 0$



$$9. f(x) = \begin{cases} x + 2, & x \leq 1 \\ x^2 + 3, & x > 1 \end{cases} \text{ at } x = 1$$

$$10. f(x) = \begin{cases} x - 3, & x \leq 0 \\ x^2 + x - 3, & x > 0 \end{cases} \text{ at } x = 0$$



Remark

If f and g are continuous at a , and if c is a constant, then the following functions are also continuous at a .

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}; \quad g(a) \neq 0$



Example

Determine the intervals within which each of the following functions is continuous.

$$1. f(x) = \frac{2x+1}{(x-1)(2x+1)}$$

$$2. f(x) = (1.12)^x$$

$$3. f(x) = \sqrt{x-7}$$

$$4. f(x) = 7x^2 - 3.2x + 0.5$$

$$5. f(x) = \frac{x+1}{(x+1)(x-3)}$$

$$6. f(x) = 4x^3 - 2x^2 + 1.3x - 5$$

$$7. f(x) = \sqrt{2x+3}$$

$$8. f(x) = e^{-0.25x}$$

$$9. f(x) = -2.1 + \ln x + 3$$

$$10. f(x) = \frac{3x-7}{2x+1}$$

$$11. f(x) = \frac{2x-3}{(2x-3)(x+4)}$$



The Derivative of a Function $f(x)$

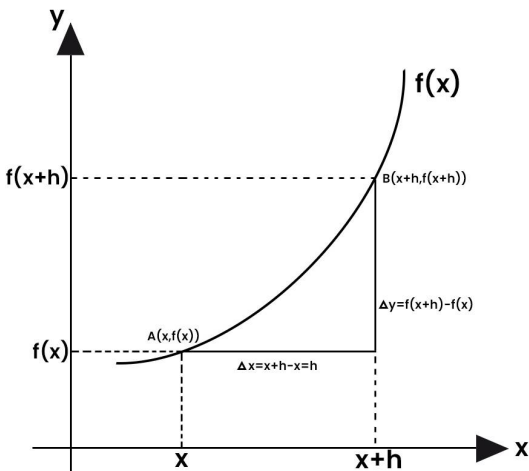


Fig 1



In Fig 1 figure above, A and B are two points on the curve $y = f(x)$. AB is a chord joining A and B . Let m denote the slope of the chord AB .

Then $m = \frac{\Delta y}{\Delta x} = \frac{f(x+h)-f(x)}{(x+h)-x}$. As Δx approaches 0, $\frac{\Delta y}{\Delta x}$ approaches a limiting value and the gradient of the chord approaches the gradient of the tangent at A . Thus the slope of the curve at A can be written as

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{(x+h) - x}$$

If this limit exists then it is called the derivative of f at x . It is denoted by $\frac{dy}{dx}$ or $(f'(x))$. This process of finding the derivative of f is called differentiating from first principles.



Remark

1. $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$
2. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Remark

1. The derivative of y with respect to x , denoted by $\frac{dy}{dx}$, is rate of change of y as a result of a change in x .
2. It is the instantaneous rate of change of y at x .
3. It is also the gradient function of the curve $y = f(x)$ at a point along the curve.



Example

Find from first principles the derivative of each of the following.

1. $f(x) = x^2$

2. $f(x) = x^2 + x$

3. $f(x) = 5x$

4. $f(x) = 8$

5. $f(x) = \frac{1}{x^2}$

Example

Differentiate the following functions from first principles.

1. $f(x) = 2x^3 + 3$

2. $f(x) = 2x^2 + 4x - 3$

3. $f(x) = 3x^2 + 1$

4. $f(x) = \sqrt{x}$

5. $f(x) = (x^2 + 3)^2$



Rules of Differentiation

1. For any constant k , if $f(x) = k$ then

$$f'(x) = \frac{d}{dx}(k) = 0$$

2. If $f(x) = x^n$, where n is any real number then

$$f'(x) = nx^{n-1}$$

3. If $y = k g(x)$, where k is any constant, then $\frac{dy}{dx} = k g'(x)$

4. If $h(x) = f(x) \pm g(x)$, where f and g are differentiable functions of x , then

$$h'(x) = f'(x) \pm g'(x)$$



Product Rule of Differentiation

If $h(x) = f(x) g(x)$, where $f(x)$ and $g(x)$ are differentiable functions of x , then

$$h'(x) = f'(x) g(x) + f(x) g'(x)$$

. Examples

1. Differentiate $h(x) = 3x^3(x^4 + 2)$.
2. Find the derivative of the following functions.
 - a. $f(x) = (2x^2 + 4x + 5)(5x - 4)$
 - b. $y = \sqrt{x}(3x^3 - 4x^2 + 8)$
 - c. $y = (6x^{4/3} + 2x)(3x^{5/3} + 4x - 1)$



Quotient Rule of Differentiation

If $h(x) = \frac{f(x)}{g(x)}$ where $g(x) \neq 0$, and f and g are differentiable functions of x then,

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{\{g(x)\}^2} \quad \text{provided } g(x) \neq 0$$

Example

Find the derivative of the following functions;

1. $f(x) = \frac{x+3}{x-2}$

2. $y = \frac{x^4-3x}{x^2+1}$

3. $y = \frac{(2x+1)(3x-2)}{x+1}$

4. $y = \frac{1+x}{1-x^2}$

5. $y = \frac{x^2-1}{x^3+1}$

6. $y = \frac{(x-3)^2}{x^2+1}$

7. $y = \frac{3x-1}{\sqrt{x^2+1}}$

8. $y = \frac{x}{1+x^2}$

9. $y = \frac{x^2-1}{x^3+1}$



Non-Differentiable Functions

In spite of all the rules of differentiation, some functions are non-differentiable i.e. some functions cannot be differentiated

at certain values. Consider $f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{|h| - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{|h|}{h}$$



Now, $\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1$ and $\lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$

Since $\lim_{h \rightarrow 0^-} \frac{|h|}{h} \neq \lim_{h \rightarrow 0^+} \frac{|h|}{h}$, $\lim_{h \rightarrow 0} \frac{|h|}{h}$ does not exist. Hence $f'(0)$ does not exist. Therefore $f(x) = |x|$ is not differentiable at $x = 0$.

Again, consider $f(x) = x^{1/3}$. $f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$. Clearly $f'(x) = x^{1/3}$ is not differentiable at $x = 0$.

The two examples above shows that a function is not differentiable at a point $x = a$ if the function is discontinuous at $x = a$.



Theorem: (Differentiability Implies Continuity)

If f is differentiable at $x = a$, then f is continuous at $x = a$

Proof

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(a) = \frac{\lim_{x \rightarrow a} f(x) - f(a)}{\lim_{x \rightarrow a} (x - a)}$$

$$f'(a) \lim_{x \rightarrow a} (x - a) = \lim_{x \rightarrow a} (f(x) - f(a))$$

$$\implies f'(a) \cdot 0 = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} f(a)$$

$$\implies 0 = \lim_{x \rightarrow a} f(x) - f(a)$$

$$\implies \lim_{x \rightarrow a} f(x) = f(a)$$

Hence f is continuous at a if f is differentiable at a .



Remark

Differentiability always implies continuity but continuity does not necessarily imply differentiability



The Chain Rule

This rule is used to differentiate composite functions that have the form $h(x) = f(g(x))$. If $y = f(u)$ and $u = g(x)$ then $y = f(g(x))$. Therefore

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

provided that $\frac{dy}{du}$ and $\frac{du}{dx}$ exists

Example: Find the derivatives of the following functions using the chain rule.

1. $f(x) = (5x^3 + 3x)^4$

2. $f(x) = (x^2 + 1)^{15}$

3. $f(x) = \sqrt[3]{2x - 4}$

4. $f(x) = \frac{5}{(2x-3)^2}$

5. $g(x) = (x + 3)^5$

6. $y = (2x^2 - 5x)^4$



The Generalised Power Rule

If u is a differentiable function of x and n is any real number with $f(x) = \{u(x)\}^n$, then

$$f'(x) = n\{u(x)\}^{n-1} \cdot u'(x)$$

. In simple terms, if $y = (ax + b)^n$, then

$$\frac{dy}{dx} = n(ax + b)^{n-1} \cdot \frac{d}{dx}(ax + b)$$

Example

Find the derivatives of the following functions

1. $y = (x^2 + 2)^5$

3. $y = (2x - 5)^{-4}$



2. $y = (3x^4 - 5)^7$

4. $y = (3x + 4)^5$

Derivatives of Logarithmic Functions

The function $f(x) = \log_e x$, where $x > 0$ is a variable and $e = 2.718$, is a constant called the natural logarithm. The natural logarithm function is commonly written as $f(x) = \ln x, x \geq 0$ or $f(x) = \ln |x|$



Properties of Natural Logarithm

If $x > 0$ and $y > 0$ then;

1. $\ln xy = \ln x + \ln y$

2. $\ln \frac{x}{y} = \ln x - \ln y$

3. $\ln x^y = y \ln x$

4. $\ln 1 = 0$

5. $\ln e = 1$

6. $e^{\ln x} = x$

7. $\ln e^x = x$



If $f(x) = \ln x$ with $x > 0$ then

$$f'(x) = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h} \text{ and thus } \frac{d}{dx}(\ln x) = \frac{1}{x}.$$

In general if

$$y = \ln f(x)$$

where f is a differentiable function of x then

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$



Example

Compute the derivatives of the following functions.

1. $f(x) = 2 \ln x$

2. $f(x) = \ln x^3$

3. $f(x) = 7 - 4 \ln x$

4. $f(x) = \ln (x^4 - 2x)$

5. $y = \ln \sqrt{6x - 1}$

6. $y = (\ln x)^3$

Example

Determine the derivative of the following functions.

1. $y = 4x^3(\ln x)$

2. $y = (\ln x)^2$

3. $y = \frac{3x^5}{\ln x}$

4. $y = 12x^3(\ln x^2)$



Derivative of the General Logarithmic Function

The general logarithmic function is the function

$$f(x) = \log_b x$$

, where $b > 0$, $b \neq 1$. The function $f(x) = \log_b x$ can be written as $f(x) = \frac{\ln x}{\ln b}$ via the change of base formula. Hence $f(x)$ becomes $f'(x) = \frac{1}{\ln b} \left(\frac{1}{x}\right)$

In general, if

$$f(x) = \log_b \{g(x)\}$$

then

$$f'(x) = \frac{1}{\ln b} \cdot \frac{g'(x)}{g(x)}$$



Example

Differentiate the following functions.

1. $y = \log_6 x$

2. $y = \log(x^3 + 9)$

3. $y = \log_5 x$

4. $y = 2x^5 \log_b x$

5. $y = \log_5(3x + 9)$

6. $y = \log_{10} \frac{x+3}{x^2+1}$

7. $y = \log_2 \frac{x^3}{x^2-1}$



Derivative of the Exponential Function

The function

$$h(x) = e^{f(x)}$$

, where $e = 2.718$ is a constant is called the exponential function. If f is a differentiable function of x , then

$$h'(x) = e^{f(x)} \cdot f'(x)$$

. Examples

Differentiate the following functions with respect to x .

1. $y = x^2 e^x$

2. $y = \frac{x^2}{e^x}$

3. $f(x) = \log_e e^x$

4. $h(x) = e^{3x-3}$

5. $g(x) = e^{6x - \frac{1}{2}x^6}$



Derivative of the General Exponential Function

The general exponential function is any function of the form

$$f(x) = a^x$$

, where $a > 0$ and $a \neq 1$.

To differentiate this function, we resort to the natural log differentiation. i.e. we take log of both sides of $y = a^x$ and differentiate it implicitly. i.e. if

$$\ln y = x \ln a \text{ then } \frac{1}{y} \frac{dy}{dx} = \ln a, \text{ thus } \frac{dy}{dx} = y \ln a$$

Therefore

$$\frac{dy}{dx} = a^x \ln a$$



Examples

Determine the derivative of the following functions.

1. $y = 10^x$

2. $y = \left(\frac{3}{4}\right)^x$

3. $y = 10^{x+3}$

4. $y = 5^{x^2}$



In general, if

$$y = a^{f(x)}$$

where f is a differentiable function of x then

$$\frac{dy}{dx} = a^{f(x)} \cdot f'(x) \ln a$$

. Proof

Let $y = a^{f(x)}$

Then $\ln y = f(x) \ln a$

$$\implies \frac{1}{y} \frac{dy}{dx} = f'(x) \ln a$$

$$\implies \frac{dy}{dx} = y f'(x) \ln a$$

$$\therefore \frac{dy}{dx} = a^{f(x)} \cdot f'(x) \ln a$$



Example

Find the derivative of these functions.

1. $y = 5^{9x-5}$

2. $y = 3^{\ln x + 5}$

3. $y = 2^{9-x}$

4. $y = 4^{\sqrt{x}}$

5. $y = 4^{x-3}$

6. $y = 5^{x^2}$



Derivative of Trigonometric Functions

1. $\frac{d}{dx}(\sin x) = \cos x$
2. $\frac{d}{dx}(\cos x) = -\sin x$
3. $\frac{d}{dx}(\tan x) = \sec^2 x$
4. $\frac{d}{dx}(\sec x) = \sec x \tan x$
5. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
6. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

In general, we have the following.

1. if $y = \sin f(x)$ then $\frac{dy}{dx} = f'(x) \cos f(x)$
2. if $y = \cos f(x)$ then $\frac{dy}{dx} = -f'(x) \sin f(x)$



Example

Differentiate the following with respect to x .

1. $y = \sin x^2$
2. $y = \sin (x^2 - 3x + 1)$
3. $y = \cos (2x^3 + x^2 - 1)$
4. $y = \cos (4x^3)$
5. $y = \sin (3x^2 + 1)$
6. $y = \tan 6x^2$



Derivative of Powers of Trigonometric functions

In general, if

$$y = \cos^n x$$

then

$$\frac{dy}{dx} = n \cos^{n-1} x \cdot \frac{d}{dx}(\cos x)$$

.

Similarly, if $y = \sin^n x$, then

$$\frac{dy}{dx} = n \sin^{n-1} x \cdot \frac{d}{dx}(\sin x)$$



Find the derivative of the following functions with respect to x

1. $y = \tan^2 x$

2. $y = 2 \cos^2 (x^2 + 1)$

3. $y = \sin^3 x$

4. $y = 3 \tan^4 x$

5. $y = x \sin 3x$

6. $y = x^3 \cos 2x$

7. $y = \cos x \sin^3 x$

8. $y = 2x^3 \cos 3x$

9. $y = x^4 \cos x$

10. $y = \cos (x^2 - x)$



Implicit Differentiation

An implicit function in two variables x and y is a function where neither variable can be expressed explicitly in terms of the other. For example, the equation $x^2 + y^2 = 3xy$ is an implicit function. Thus far, all the functions that we have studied have the form $y = f(x)$. Algebraically, this means that we can solve for y explicitly in terms of x such as $y = x^3 + 3x^2 - 1$. However, not all equations are expressed in this form. For example, for the equation, $xy^4 + y^2 - xy + 3x^2 - 5 = 0$, solving for y explicitly will be difficult. How then do we find the derivative $\frac{dy}{dx}$ in this case? The key is to use implicit differentiation. To differentiate a function implicitly, we follow the steps below.



1. Differentiate the function term by term to give an equation in $\frac{dy}{dx}$, x and y .
2. Make $\frac{dy}{dx}$ the subject of the equation.
3. The following are worth noting.
 - a. $\frac{d}{dx}(y) = \frac{dy}{dx}$
 - b. $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$
 - c. $\frac{d}{dx}(2y^3) = 6y^2 \frac{dy}{dx}$
 - d. In general $\frac{d}{dx}(f(y)) = f'(y) \frac{dy}{dx}$, where $f'(y)$ is the derivative of $f(y)$ with respect to y .
4. Terms such as $3x^2y$, xy , $x \frac{dy}{dx}$ should be treated as products, and $\frac{x^2}{y}$, $\frac{2x}{y}$ should be treated as quotients.



Example

Use implicit differentiation to determine $\frac{dy}{dx}$ for the following functions

1. $y^3 - 4x^2 - 7 = 0$

2. $x^{1/3} + y^{1/3} - 10 = 0$

3. $4e^y = x^2$

4. $2x + 1 = \sqrt{2 - y^2}$

5. $x^2y^2 - 7x^3 - 5 = 0$

6. $x^2 + y^2 = 25$

7. $3xy + 6y^2 = 15$

8. $y + xy + y^2 = 2$



Higher Order Derivatives

So far we have considered the rate of change of y with respect to x when $y = f(x)$. This is called the first derivative. The result of differentiating $\frac{dy}{dx}$ with respect to x is called the second derivative of y with respect to x . This is denoted by

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

If the second derivative is also differentiated with respect to x the third derivative $\frac{d^3y}{dx^3}$ is obtained and so on.

Remark

If $y = f(x)$ then $\frac{dy}{dx} = f'(x)$, $\frac{d^2y}{dx^2} = f''(x)$, $\frac{d^3y}{dx^3} = f'''(x)$ etc.



Example

Find $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$ for each of the following functions.

1. $y = 3x^4 + 2x^3 + \frac{2}{x^2}$

2. $y = x^6 + 2x^3 - 3x^2 + 4x + 1$

3. $y = x^4 - x^3 + 4x - 1$

4. $y = 3x^3(x^2 + 4)^2$

5. $y = -5 \cos 3x$

6. $y = \frac{x}{1+x^2}$



The Derivative of a Function with Parametric Representation

Consider the function $y = f(x)$ with parametric representation $x = f(t)$ and $y = f(t)$, where t is the parameter. In such cases the gradient function $\frac{dy}{dx}$ is given in terms of the variable parameter using the chain rule of differentiation i.e.

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

. Now by virtue of the fact that x is a function of t , we can only find $\frac{dx}{dt}$ and write $\frac{dt}{dx}$ as

$$\frac{dt}{dx} = \frac{1}{\frac{dx}{dt}}$$



Example

1. If $x = t^2$ and $y = t^3$, find $\frac{dy}{dx}$ in terms of t and show that the above parametric representation is of the curve $y = x^{3/2}$.
2. Find $\frac{dy}{dx}$ in terms of t when $x = at^2$ and $y = 2at$, where t is a parameter and a is a constant, Find the equation with this parametric representation.
3. Find $\frac{dy}{dx}$ in terms of t in each of the following cases
 - a. $x = \frac{t}{1-t}, \quad y = \frac{t^2}{1-t}$
 - b. $x = \frac{2t}{t+2}, \quad y = \frac{3t}{t+3}$
4. Find the curve whose parametric representation is given in each case in (3) above.



Examples

1. A cylindrical container of thin metal is to hold 10cm^3 of water. If an engineer wishes to use the least material of metal for the construction of the cylinder, show that $A = 2\pi r^2 + \frac{20}{r}$, where r is the radius of the base area, h is the height of the cylinder and A is the surface area. Hence show that the area of the metal needed is minimum when the height of the cylinder is twice the base radius.
2. A point P whose x -coordinate is α is on the line $y = 3x - 7$. If Q is the point $(4, 1)$, show that $|PQ|^2 = 10\alpha^2 - 56\alpha + 180$, and find the value of α which makes $|PQ|^2$ minimum.



3. A sealed cylindrical can is of radius r and height h . The area of its total surface is A and its volume is V
- Find an expression for A in terms of r and h .
 - If $A = 24\pi$, find an expression for h in terms of r .
Hence find an expression for V in terms of r .
 - Find the value of r which maximises V .
4. A cylinder is such that the sum of its height and the circumference of its base is 5cm . Express the volume V in terms of the radius of the base r . What is the maximum value of the cylinder?



Applications of Differentiation

1. The gradient Function: If $y = f(x)$ then $\frac{dy}{dx}$ or $f'(x)$ is called the gradient function. The gradient of a curve at a point $P(a, b)$ is obtained by substituting the values a and b into the expression for $\frac{dy}{dx}$. This value $\frac{dy}{dx}|_{(a,b)}$ is the same as the gradient of the tangent to the curve at the point $P(a, b)$



Example

Find the gradient of the curve at the indicated point.

1. $f(x) = (x^2 + 3)^2$ at $x = \frac{1}{2}$

2. $y = (x - 3)(x^2 + 2)$ at, $x = 1$

3. $x^2 + y^2 = 9$ at $x = 1$

4. $y = x^2 - 3$ at $x = 1$

5. $y = 3x^2 - 2$ at $x = 3$

6. $y = \sqrt{x}$ at $x = 2$

7. $y = (x - 3)(x + 4)$ at $x = -1$

8. $y = \frac{1}{x}$ at $x = 3$



Examples

1. Find the coordinates of the point on the curve $y = \frac{2}{x^2}$ where its gradient is $\frac{1}{2}$
2. Find the values of x for which the gradient function of the curve $y = \frac{1}{4}x^4 + \frac{2}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$ is zero
3. Find the coordinate of the point on the given curve where its gradient, m has the given value.
 - (a) $y = x^2 - x + 3$; $m = 1$
 - (b) $y = x^3 + x^2$; $m = 1$
 - (c) $y = 5 + 3x - 2x^2$; $m = -3$
 - (d) $y = (x + 3)(x - 5)$; $m = 0$
 - (e) $y = (x + 1)(x - 1)$; $m = 2$



2. Equations of Tangents and Normals to a Curve:

- a. If $y = f(x)$ then $\frac{dy}{dx}|_{(a,b)}$ is the gradient of the tangent to the curve at $P(a, b)$. If $m = \frac{dy}{dx}|_{(a,b)}$ then the equation of the tangent to the curve at $P(a, b)$ is

$$y - b = m(x - a).$$

- b. The normal to a curve at the point of contact is the straight line perpendicular to a tangent to the curve at that point of tangency. If m_1 is the gradient of the tangent to the curve and m_2 is the gradient of the normal then

$$m_1 \cdot m_2 = -1$$



Example

Given that $y = x^3 - 4x^2 + 5x - 2$ and that P is a point on the curve at $x = 3$,

- (a) find $\frac{dy}{dx}$,
- (b) find the coordinates of P .
- (c) calculate the gradient of a tangent at P
- (d) find the equation of the tangent at P
- (e) find the equation of the normal at P
- (f) find the value of x for which the curve has gradient 5



Example

Find the equation of the tangent and the normal to the curve $y = x^3 - 3x + 2$ at the point $P(2, 4)$.

Example

Find the coordinates of the points on the curve $y = x^3 - 4x^2 - 3x - 2$ where the tangent is parallel to the x-axis.

Example

The gradient function of a curve at a point $P(x, y)$ is $\frac{dy}{dx} = 4x^2 - 3x + 1$. If the curve passes through the point $A(1, 0)$, find the equation of the tangent to the curve at A .



Example

Given that $y = (x - 2)(x^2 - 3x - 5)$, find $\frac{dy}{dx}$. Determine the gradient of the tangent to the curve at the point where it cuts the x -axis.

Example

Find the gradient and the equation of a tangent and a normal to the curve $y = 5x^3 - 7x^2 + 3x + 2$ at $(1, 3)$.

Example

A tangent to the curve $y = x^3 + kx$, where k is a constant, at $x = 1$ passes through the points $A(-1, 6)$ and $B(2, -15)$. Find the value(s) of k .

Example

Find the gradient of $\frac{x^2}{25} + \frac{y^2}{4} = 1$ at $P(3, \frac{8}{5})$.



3. Rates of Change: Suppose we have some quantity y which depends on x and also varies with x . The rate of change of y with respect to x is given by $\frac{dy}{dx}$. For instance, if V is the volume of a sphere and r is its radius then $\frac{dV}{dr}$ is the rate of change of V with respect to r . If the quantity y is increasing with respect to x then $\frac{dy}{dx} > 0$ and if y is decreasing with to x then $\frac{dy}{dx} < 0$.

Remark

Rates of change are by convention with respect to time.



Example

A circular ink blot has a radius 2cm . At what rate is the area increasing with respect to the radius?

Example

A spherical balloon is being inflated and the volume is increasing at the rate of $15\text{cm}^3/\text{sec}$. At what rate is the radius increasing when it is 10cm ?



Example

A spherical balloon is blown up so that its volume increases at the rate of $2\text{cm}^3/\text{s}$. Find the rate of increase of the radius when the volume of the balloon is 50cm^3 .

$$[\text{Hint} : V_s = \frac{4}{3}\pi r^3, \quad S_A = 4\pi r^2]$$

Example

A spherical bubble is decreasing in volume at a rate of $2\text{cm}^3/\text{s}$. Find the rate at which the surface area is diminishing when the radius is 3cm . $[\text{Hint} : V_S = \frac{4}{3}\pi r^3, \quad S_A = 4\pi r^2]$



Examples

1. A spherical balloon is inflated by a gas being pumped in at a constant rate of $200\text{cm}^3/\text{s}$. What is the rate of increase of the surface area of the balloon when the radius is 100cm ?
2. An elastic spherical balloon is being blown up so that the radius is increasing at a rate of $1\text{cm}/\text{s}$. Calculate the rate at which the volume of the balloon is increasing when the radius is 5cm .
3. The side of a cube is increasing at a rate of $60\text{m}/\text{s}$. Find the rate of increase of the volume when the length of the side is 9cm .
4. The volume of a cube is increasing at a rate of $\frac{5}{2}\text{cm}^3\text{s}^{-1}$. Find the rate of change of the side of the base when its length is 2cm .



4. Small Changes: From previous discussions, we learnt that

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}.$$

This means that $\frac{\Delta y}{\Delta x} \simeq \frac{dy}{dx}$ when Δx is so small. Hence

$$\Delta y \simeq \frac{dy}{dx} \cdot \Delta x.$$

This approximation can be used to estimate the small change Δy in y if $\frac{dy}{dx}$ can be found when the small change Δx in x is given. Thus the change in y is given by

$$\Delta y = \frac{dy}{dx} \cdot \Delta x.$$



We can also use this approximation to estimate percentage changes. In general, if x is increased by $p\%$, then

$$\Delta x = \frac{p}{100} * x$$

and the approximate percentage increase in y is

$$\frac{\Delta y}{y} * 100$$

.



Examples

1. The surface area of a sphere is $4\pi r^2$. If the radius of the sphere is increased from 10cm to 10.1cm , what is the approximate increase in the surface area?
2. The radius r of a circle is 5cm . Find the increase in the area A of the circle when the radius expands by 0.01cm
3. The side of a cube increases by 5% . Find the corresponding percentage increase in the volume
4. Find the approximate value for the square root of 16.01
[Let $y = \sqrt{x}$, $x = 16$, $\Delta x = 0.1$]
5. The radius of a sphere is increased by 2.5% . Find the percentage increase in the volume of the sphere
6. If $y = \sqrt{x}$, find the approximate increase in y if x is increased from 4 to 4.01



5. Maxima and Minima:

1. Stationary Points: A point on a curve at where $\frac{dy}{dx} = 0$ is called a stationary point and the value of the function at this point is called the stationary value. At the stationary point, the tangent to the curve is parallel to the x-axis. To find the stationary point, we put $\frac{dy}{dx} = 0$ and solve for the resulting equation. Stationary points are also called turning points.



2. Procedure for testing and distinguishing between stationary points:

- Given $y = f(x)$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$
- Put $\frac{dy}{dx} = 0$ and solve the resulting equation for the turning value.
- Substitute the value of x into the original equation $y = f(x)$ to find the corresponding y coordinate value.
This establishes the coordinate of the stationary point
- Substitute the value(s) of x found in step (b.) into $\frac{d^2y}{dx^2}$.
If $\frac{d^2y}{dx^2} < 0$ then the point is a maximum point. Finally, If $\frac{d^2y}{dx^2} > 0$ then the point is a minimum point.



Examples

1. A curve is defined by $f(x) = x^3 - 6x^2 - 15x - 1$. Find;
 - a. the derivative of f
 - b. the gradient of the curve at the point where $x = 1$
 - c. the maximum and the minimum points
 - d. the maximum and the minimum values of f
2. Find the minimum/maximum point of $y = x^2 - 3x + 5$.
3. Find the stationary points of $y = \frac{1}{3}x^3 - 2x^2 + 3x$ and identify their nature.
4. The curve $y = x^2 - ax + b$ has a turning point at $P(1, 3)$. Find the values of a and b .
5. The function $f(x) = ax^2 + bx + c$ has gradient function $4x + 2$ and stationary value of 1. Find the values of a , b and c .



Examples

1. A rectangular sheet of metal 60cm by 50cm has squares each of side $x\text{ cm}$ cut off from its corners. The remainder is then formed into a cuboid of volume $V\text{ cm}^3$. Find the value of x for which the volume is maximum. Hence find the maximum volume.
2. The length of the sides of a rectangular sheet of metal are 8cm and 3cm . A square of side $x\text{ cm}$ is cut off from each corner of the sheet and the remaining piece is folded to make an open box.
 - (a) Show that the volume, V of the box is given by
$$V = 4x^3 - 22x^2 + 24x\text{ cm}^3.$$
 - (b) Find the value of x for which the volume is maximum and find the maximum volume.



3. A rectangular box with a square base $x \text{ cm}$ and height $h \text{ cm}$ is to be constructed from a cardboard,
- If the box holds a volume of 200cm^3 , find a formula for h in terms of x .
 - Show that the area of the cardboard needed to make the box is given by $2x^2 + \frac{800}{x}$
 - Find, to 1 d.p the least area of cardboard needed to make the box.
4. A rectangle has perimeter 28m . What is its maximum area?
5. 100cm of fencing is to be used to make a rectangular enclosure. Find the greatest possible area that can be enclosed.



6. A ball is thrown vertically upwards from the ground level and its height after t seconds is $h = (50t - 16t^2) m$. Find,
- The greatest height reached.
 - The time taken to reach the greatest height
7. Determine the area of the largest piece of rectangular ground that can be enclosed by $100m$ of fencing, if part of an existing straight wall is used as one side.
8. A cylindrical can has a radius r and a height h . The sum of the radius and the height is $2m$. Find an expression for the volume, V of the cylinder in terms of r only. Hence find the maximum volume



6. Increasing and Decreasing Functions: The derivative of a function $y = f(x)$; $\frac{dy}{dx}$ gives the gradient function of the curve. If $\frac{dy}{dx} > 0$ then the function is increasing and decreasing when $\frac{dy}{dx} < 0$.

Example

Find the range of values of x for which the following functions are increasing.

1. $x^2 - 2x - 5$
2. $2x^3 + 3x^2 - 12x + 5$
3. $x^3 - 3x^2 + 3x + 2$



Example

Find the interval on which $f(x) = x^3 + x^2 - 5x - 5$ is

1. Increasing
2. Decreasing

Example

Find the range of values of x for which the function

$$f(x) = x^3 - 12x + 5 \text{ is}$$

1. Increasing
2. Decreasing



7. Curve Sketching: Let $y = f(x)$ be a function that represents a curve we wish to sketch. The steps below are used to sketch any curve whose equation is given.

Step 1: Find the intercepts on the x and y axes

1. For intercept on x -axis, put $y = 0$ and solve for x
2. For intercept on y -axis, put $x = 0$ and solve for y

Step 2: Find the turning points. At the turning points put $\frac{dy}{dx} = f'(x) = 0$ and solve the resulting equation for x . Substitute these value(s) of x into the original equation $y = f(x)$ to find the corresponding y -coordinate(s). This establishes the coordinates of the turning point(s).



Step 3: Test whether the turning points are maximum/minimum. Use the following conditions.

1. If $\frac{d^2y}{dx^2} > 0$ at the turning points then the turning points are minimum.
2. If $\frac{d^2y}{dx^2} < 0$ at the turning points then the turning points are maximum.
3. If $\frac{d^2y}{dx^2} = 0$ at the turning points then the turning points are inflection points.



Example

1. Sketch the curve $y = x^3 - 7x^2 + 15x - 9$ indicating clearly its points of intersection with the coordinate axes and its turning points.
2. Sketch the following curves.
 - a. $y = x(x^2 - 16)$
 - b. $y = x^2 - 4x - 5$
 - c. $y = x^2 - 6x$
 - d. $y = 5x - x^2$
3. A function f is defined by $f(x) = 2x^3 + 3x^2$.
 - a. Find the turning points and distinguish between them.
 - b. Sketch the curve of $f(x)$.
4. A function f is defined by $f(x) = \frac{x^3}{3} - \frac{x^2}{2} - 6x + \frac{5}{3}$.
 - a. Find the turning points and distinguish between them.
 - b. Sketch the curve of $f(x)$.



Applications of Differentiation to Linear Kinematics

Kinematics is the study of displacements, velocity, and acceleration. Consider a particle in motion in a straight line such that its displacement from a fixed point is x meters after time t seconds. Then displacement is a function of time i.e. $x = f(t)$. The S.I unit of displacement is cm , m or km

Velocity: It is the rate of change of displacement with time i.e. $v = \frac{dx}{dt}$. The S.I unit of velocity is cms^{-1} , ms^{-1} , or kmh^{-1} . If $v = 0$ then the particle is momentarily at rest. If $v < 0$ then the particle is moving in the opposite direction to its initial direction.



Acceleration: Acceleration is the rate of change of velocity with time i.e. $a = \frac{dv}{dt}$. The S.I unit of acceleration is cms^{-2} , ms^{-2} , or kmh^{-2} . If $a = 0$ then the velocity of the particle is constant. If $a > 0$, the particle is accelerating and it is decelerating/retarding when $a < 0$.

Average Velocity: If a particle moves a distance x_1 in time t_1 and another distance x_2 in time t_2 , where $t_2 > t_1$, then the average velocity of the particle within the time interval t_1 and t_2 is given by;

$$\text{Average Velocity} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$



Average Acceleration: If the velocity of a particle at t_1 is v_1 and at t_2 is v_2 , where $t_2 > t_1$, then the average acceleration of the particle within the time interval t_1 and t_2 is given by;

$$\text{Average Acceleration} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$



Examples

1. A particle moves in a straight line such that its distance x meters from a fixed point 0 after t seconds is given by $x(t) = t^3 - 2t^2 + t + 10$. Calculate
 - (a) its average velocity after 2 seconds,
 - (b) its average velocity during the 3rd second,
 - (c) the average acceleration from $t = 1$ to $t = 3$.
 - (d) When and where is the particle momentarily at rest?
2. An object is thrown into the air such that its height, h meters after t seconds is $h(t) = 36t - 4t^2$. Find
 - a. the velocity at $t = 2$,
 - b. the time taken to reach the maximum height.



3. An object projected vertically upwards satisfies the equation $h(t) = 27t - 3t^2$, where h meters is the height after t seconds.
- Find the time it takes to reach the highest point.
 - How high does it go?
4. A ball thrown vertically upwards at time t has height $h(t) = 32t - 8t^2$.
- Find the velocity of the ball at $t = 1$ second
 - Find the max height it reached.
5. A particle p moves in a straight line so that its distance from a fixed point O after t seconds is $x(t) = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$. Show that the particle is at rest at two different times and find these times. Find the acceleration of the particle at these times and interpret your results.



6. A particle moves in a straight line so that its distance x from a fixed point 0 after t seconds where $t \geq 0$ is given by $x(t) = 9t^2 - 2t^3$.
- (a) Find v and a at $t = 3$.
 - (b) Find x when $t = 4$ and show that the motion is towards the fixed point 0.
7. A particle projected from a fixed point with velocity 10ms^{-1} , moves in a straight line such that its velocity after t seconds is $v(t) = bt^2 + 7t + c$, where b and c are constants. 1 second after projection, its acceleration is 1ms^{-1} . Find the value of b and c .



8. A particle moves along a straight line such that its distance X from a fixed point after t seconds is
- $$X(t) = -\frac{1}{3}t^3 + t^2 - \frac{5}{7}$$
- (a) Find the time when the particle is momentarily at rest.
(b) Find the average acceleration of the particle between $t = 0$ and $t = 10$.
9. A particle p moves along the x-axis such that its distance x from a fixed point 0 is given by $x(t) = t^3 - 6t^2 + 9t$. Calculate
- (a) the times at which the particle is stationary.
(b) the distance of p at these times.
(c) the acceleration of p at these times.
(d) the velocity of the particle when $a = 0$.



10. A missile fired from the ground level rises x meters vertically upwards in t seconds and $x(t) = 100t - \frac{25}{2}t^2$. Find
- (a) the initial velocity of the missile.
 - (b) the time when the height of the missile is maximum.
 - (c) the maximum height reached.
 - (d) the velocity with which the missile strikes the ground.
11. A particle p moves in a straight line such that displacement from a fixed point A is $x(t) = (3t^2 + 4)$
- (a) Find the velocity of the particle at $t = 2$.
 - (b) Find the initial displacement of p from A .
12. A particle starts from rest and travels in a straight line so that its acceleration is $(t + 3)ms^{-2}$ at time t . Find the distance traveled in the interval of the time from $t = 1$ to $t = 3$.



Application of Differentiation to Business and Economics

Marginal Analysis: Many decisions made by managers in business involve analyzing the effect on the dependent variable when a small change is made to a specific independent value. For example, a firm may wish to consider changing the price of an item and examining how this change affects the revenue or profit of the product. Marginal analysis can be defined as the study of the amount of change in the dependent variable that results from a change in an independent variable. A unit change means a change of one single unit. This change in the dependent variable is a direct application of the derivative.



The price-demand function: The price-demand function, P is the price $P(x)$ at which exactly x units of the product is bought.

The cost function: The cost of producing x units of the product with variable cost a and fixed cost b is given by the cost function $C(x) = ax + b$ i.e.

$$TC = VC + FC.$$

It should be noted that since variable costs are often expressed as a function, $C(x)$ may be a higher-order polynomial function.

The total revenue function: The total revenue function generated by producing and selling x units of the product at a price P is given by $R(x) = xP(x)$.



The profit function: The profit function π generated after producing and selling x units of the product is given by $\pi(x) = R(x) - C(x)$ i.e. Profit = $TR - TC$

Marginal Cost Function: The marginal cost function $MC(x)$ is the approximate cost of producing one additional unit at a production level x . It is the derivative of total cost i.e. $MC(x) = \frac{d}{dx}C(x)$

Marginal Revenue Function: The marginal revenue function $MR(x)$ is the approximate gain in revenue by producing one additional unit at a production level. It is the derivative of the total revenue i.e. $MR(x) = \frac{d}{dx}(R(x))$.



Marginal Profit Function: The marginal profit function $MP(x)$ is the approximate gain in profit by producing one additional unit at a production level x . It is the derivative of profit function i.e. $MP(x) = \frac{d}{dx}(\pi(x))$

Average Cost Function: The average cost function is the per unit cost of producing x items. It is given by

$$AC(x) = \frac{C(x)}{x} = \frac{\text{Total cost}}{\text{No. of units produced}}$$



Average Revenue Function: This is the per unit revenue generated by producing and selling x items. It is given by

$$AR(x) = \frac{R(x)}{x} = \frac{\text{Total revenue}}{\text{No. of units sold}}$$

Average Profit Function: The average profit function is the per unit profit of producing and selling x items. It is given by

$$AP(x) = \frac{\pi(x)}{x} = \frac{\text{Total profit}}{\text{No. of units sold}}$$



Conditions for Minimising Average Cost

$$\text{Now, } AC(x) = \frac{C(x)}{x}$$

$$\implies \frac{d}{dx}(AC) = \frac{x \frac{d}{dx}(C(x)) - C(x) \frac{d}{dx}(x)}{x^2}$$

$$\implies \frac{d}{dx}(AC) = \frac{xC'(x) - C(x)}{x^2}$$

$$\text{For minimum average cost } \frac{d}{dx}(AC) = 0$$

$$\therefore \frac{xC'(x) - C(x)}{x^2} = 0$$

$$\implies xC'(x) - C(x) = 0$$

$$\implies C'(x) = \frac{C(x)}{x} \dots \dots \dots (1)$$

Equation (1) says that average cost is minimum when marginal cost is equal to average cost



Example

A company estimates that the cost of producing x items of a commodity is given by $C(x) = 2680 + 2x + 0.001x^2$.

1. Find the cost of producing 100 items.
2. Find the average cost of producing 200 items.
3. Find the marginal cost of producing 300 items.
4. At what production level will the average cost be minimum? What is the minimum average cost?



Conditions for Maximum Profit

If x units are sold then total profit is $\pi(x) = R(x) - C(x)$.

$$\implies \pi'(x) = R'(x) - C'(x) \dots\dots\dots(*)$$

For maximum product $\pi'(x) = 0$

$$\implies R'(x) = C'(x) \dots\dots\dots(2)$$

Equation (2) says that for profit to be maximum, marginal revenue should be equal to marginal cost.



From (*), $\pi''(x) = R''(x) - C''(x)$.

But for max profit, $\pi''(x) < 0$.

$$\text{Therefore } R''(x) - C''(x) < 0$$

$$\implies R''(x) < C''(x) \dots\dots\dots(3)$$

(3) says that the rate of increase of marginal revenue should be less than rate of increase of marginal cost for profit to be maximum.



Examples

1. Determine the production level that will maximize the profit for a company with respective cost and dd function $C(x) = 84 + 1.26x - 0.01x^2 + 0.00001x^3$ and $P(x) = 3.5 - 0.01x$.
2. The average cost of producing x units of a commodity is $AC(x)=21.4-0.002x$. Find the marginal cost at a production level of 1000 units and interpret your answer.



3. For each cost function given below, find;
- the cost, average cost and the marginal cost at a production level of 1000 units.
 - the production level that will minimize the average cost.
 - The minimum average cost.
 - $C(x) = 40000 + 300x + x^2$
 - $C(x) = 25000 + 120x + 0.1x^2$
 - $C(x) = 16000 + 200x + 4x^{3/2}$
 - $C(x) = 10000 + 340x - 0.3x^2 + 0.0001x^3$
4. For the given cost and dd function, find the production level that will maximize profit.
- $C(x) = 680 + 4x + 0.01x^2$; $P(x) = 12$
 - $C(x) = 680 + 4x + 0.01x^2$; $P(x) = 12 - \frac{x}{500}$
 - $C(x) = 1450 + 36x - x^2 + 0.001x^3$; $P(x) = 60 - 0.01x$
 - $C(x) = 16000 + 500 - 1.61x^2 + 0.004x^3$;
 $P(x) = 1700 - 7x$



Example

DBS Industry has determined that its cost of producing x IBR brand of roofing sheets can be modeled by the function

$C(x) = 2x^2 + 15x + 1500$, $0 \leq x \leq 200$, where x is the number of IBR roofing sheets produced each week and $C(x)$ represents the weekly cost in Ghana cedis. The company also determines that the price -dd function for the roofing sheets is $P(x) = -0.3x + 460$.

1. Determine the profit function π for the roofing sheets.
2. Determine the marginal profit function.
3. Compute $MP(60)$ and $MP(145)$ and interpret the results



Example

The revenue function for the production of x boxes of roofing nails per week is $R(x) = 0.3x^2 + 460x$. Compute $R(110)$ and $MR(110)$ and interpret them.

Example

Odike shoe manufacturing company knows that, for its executive shoes, the daily cost function can be modeled by the function $C(x) = 700\sqrt{x} + 500$, $0 \leq x \leq 500$, where x is the number of pairs of shoes produced daily and $C(x)$ is the daily cost in cedis.

1. Determine the average cost function, $AC(x)$.
2. Evaluate and interpret $C(400)$ and $AC(400)$.



Example

Determine the marginal cost function $MC(x)$ and marginal profit function $M\pi(x)$ for each of the following;

1. $C(x) = 5x + 500$; $P(x) = 6$
2. $C(x) = 12x + 4500$; $P(x) = 15$
3. $C(x) = \frac{x^2}{100} + 7x + 1000$; $P(x) = -\frac{x}{20} + 15$
4. $C(x) = \frac{1}{100}x^2 + \frac{1}{2}x + 8$; $P(x) = 1 - \frac{x}{200}$



Integration

Until now, we have focused almost exclusively on how to determine the derivative of a function $y = f(x)$ and interpret the resulting rate of change function. But what happens if we are given the rate of change function and we wish to find the original function? In this section, we will study the basis of integration, a process that reverses the process of differentiation and finds the original function when its derivative is given.



The Indefinite Integral

If $F'(x) = f(x)$, then

$$\int f(x) = F(x) + c.$$

In this notation,

- $\int$ is the integral sign
- $f(x)$ is the integrand.
- $F(x)$ is the integral of $f(x)$.
- c is a constant of integration.
- dx tells us the variable of integration.



Rules of Integration

1. **The Power Rule:** For any real number n , where $n \neq -1$, the indefinite integral of x^n is $\int x^n dx = \frac{1}{n+1}x^{n+1} + c$

Example

Compute the following integrals.

1. $\int x^8 dx$

2. $\int \sqrt{x} dx$

3. $\int x^{-5} dx$

4. $\int x^{1/3} dx$



2. **Constant Rule:** If k is any constant, then the indefinite integral of k with respect to x is $\int k \, dx = k \int dx = kx + c$

Example

Evaluate the following integrals.

1. $\int (2x - 5) \, dx$

2. $\int 3x^2 \, dx$

3. $\int 2x^4 \, dx$



3. Sum and Difference Rule: For integrable functions $f(x)$ and $g(x)$ we have

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

Example

Find the following integrals

1. $\int (x^2 + 3) dx$
2. $\int (x^{1/3} + 5) dx$
3. $\int \frac{2x^3 - 3x}{4x} dx$
4. $\int (x^3 + \frac{1}{x^2} + \sqrt{x}) dx$
5. $\int (2x^2 - \frac{1}{x^2} + x) dx$



Definite Integrals

Integrals containing an arbitrary constant c in their results are called indefinite integrals since their precise value cannot be determined without further information.

Definite integrals are those with limit of integration. The definite integral of $f(x)$ from a to b is given by

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a),$$

whenever $F'(x) = f(x)$.



Example

Evaluate the following integrals with respect to x

1. $\int_1^3 (3x^2 + 2x + 4) dx$

2. $\int_1^2 \frac{(x+1)(x^2-2x+2)}{x^2} dx$

3. $\int_{-2}^2 (x^3 - x^2 + 1) dx$

4. $\int_{-2}^2 (x^2 - 2)(2x - 1) dx$

Example

If $\int_{-1}^a 2(x + 2) dx = 15$, where $a > 0$, find the value(s) of a .



Integration of Simple Trigonometric Functions

$$1. \int \cos x \, dx = \sin x + c$$

$$2. \int \sin x \, dx = -\cos x + c$$

$$3. \int \tan x \, dx = \ln \sec x + c$$

$$4. \int \sec^2 x \, dx = \tan x + c$$

$$5. \int \operatorname{cosec}^2 x \, dx = -\cot x + c$$

$$6. \int \sec x \tan x \, dx = \sec x + c$$

$$7. \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$$



Integration by Substitution

Functions which are to be integrated are not always in the standard form which can easily be integrated. However, it is possible to change them into a form which can be integrated by using an algebraic substitution.

Example

Evaluate the following integrals

1. $\int x\sqrt{1-x^2} dx$

2. $\int 3x(4x^2+3)^5 dx$

3. $\int \frac{2x}{\sqrt{4x^2-1}} dx$

4. $\int x(x^2-3)^4 dx$

5. $\int x^2(1-x^3) dx$

6. $\int (2+7x)^7 dx$

7. $\int \frac{1}{(4x+5)^3} dx$

8. $\int \frac{1}{\sqrt{1-2x}} dx$



Example

Compute the following definite integrals

$$1. \int_0^2 \frac{3x}{\sqrt{2x^2+1}} dx$$

$$2. \int_1^3 5x\sqrt{(2x^2+7)} dx$$

$$3. \int_0^1 x^2\sqrt{x^3+1} dx$$

$$4. \int_0^2 x(x^2+1)^3 dx$$

$$5. \int_{-1}^2 \frac{3}{(2y+3)^{1/3}} dy$$

$$6. \int_2^{10} \frac{1}{\sqrt{2y+5}} dy$$



Integration by Partial Fraction

Definition: A rational function is any function of the form

$$\frac{P(x)}{Q(x)},$$

where $P(x)$ and $Q(x)$ are polynomials in x and $Q(x) \neq 0$. For example,

$$f(x) = \frac{x+1}{x^2-1}$$

and

$$g(x) = \frac{x^2 - 2x - 8}{x+4}$$

are rational functions.



If the degree of $P(x)$ is less than that of $Q(x)$ then $\frac{P(x)}{Q(x)}$ is a proper rational function. Example $f(x) = \frac{x+1}{x^2-1}$ is a proper rational function.

If the degree of $P(x)$ is greater than or equal to the degree of $Q(x)$ then $\frac{P(x)}{Q(x)}$ is an improper rational function. Example $g(x) = \frac{x^2-2x-8}{x+4}$ is an improper rational function.



Resolution of Rational Functions into Partial Fractions

A rational function which may be expressed as a sum of separate fractions is said to be resolved into partial fractions. For example,

$$\frac{1}{x-2} + \frac{3}{x+1} = \frac{4x-5}{x^2-x-2}.$$

The reverse process of splitting $\frac{4x-5}{x^2-x-2}$ into its component fractions $\frac{1}{x-2} + \frac{3}{x+1}$ is called resolving into partial fractions



Type I - Denominator with Linear Factors

Example

Express $\frac{6}{(x+3)(x-3)}$ as a partial fraction.

$$\frac{6}{(x+3)(x-3)} = \frac{A}{(x+3)} + \frac{B}{(x-3)}.$$

Example

Resolve the following rational functions into partial fractions.

1. $\frac{3x+2}{x^2+x-2}$
2. $\frac{11-3x}{x^2-2x-3}$
3. $\frac{9}{2x^2+x}$
4. $\frac{4x-2}{x^2+2x}$



Type II - Denominator with Quadratic Factors

Example

Resolve the rational function $\frac{11-4x-x^2}{(x+2)(x^2+1)}$ into partial fractions.

$$\frac{11 - 4x - x^2}{(x + 2)(x^2 + 1)} = \frac{A}{(x + 2)} + \frac{Bx + C}{(x^2 + 1)}$$

Examples

Express each of the following in partial fractions.

1. $\frac{3x+1}{(x-1)(x^2+1)}$

2. $\frac{7x^2+5x+13}{(x^2+2)(x+1)}$

3. $\frac{x^2+1}{(x^2+2)(x-1)}$

4. $\frac{1}{x(x^2+1)}$

5. $\frac{13x}{(x-3)(x^2+x+1)}$

6. $\frac{x^2+9x+6}{(x^2+1)(x+1)}$

7. $\frac{x}{(x-1)(x^2+1)}$

8. $\frac{11-4x-x^2}{(x+2)(x^2+1)}$



Type III - Denominator with Repeated Factor

Example

Express $\frac{1}{(x+2)(x-1)^2}$ as a partial fraction.

$$\frac{1}{(x+2)(x-1)^2} = \frac{A}{(x+2)} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}$$

Example

Express these rational functions as a partial fraction.

1. $\frac{1}{(x+2)(x-1)^2}$

2. $\frac{5x^2-2x-19}{(x+3)(x-1)^2}$

3. $\frac{2}{x^2(x-1)}$

4. $\frac{5x+4}{(1-x)(x+2)^2}$

5. $\frac{11x+12}{(x+2)(x-3)^2}$

6. $\frac{x^2-1}{x^2(2x+1)}$



Type IV - Improper Fractions

Example

Resolve $\frac{x^2+1}{x^2-3x+2}$ into partial fractions.

$$\frac{x^2 + 1}{x^2 - 3x + 2} = 1 + \frac{3x - 1}{x^2 - 3x + 2}$$

Example

Resolve these rational functions into partial fractions.

1. $\frac{2x^2+1}{(x-1)(x+2)}$

2. $\frac{x^2+1}{x^2-3x+2}$

3. $\frac{x^3+3}{(x+1)(x-1)}$

4. $\frac{x^2-2}{(x+3)(x-1)}$

5. $\frac{x^2+3}{(x-2)(x+2)}$



Integrals that yield Logarithmic Functions

We recall that $\frac{d}{dx}(\ln x) = \frac{1}{x}$, and in general if f is a differentiable function of x then $\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \cdot f'(x)$.

Therefore

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

Also, the power rule for integration states that

$$\int x^n dx = \frac{1}{n+1} \cdot x^{n+1} + c, \text{ provided } n \neq -1.$$

To find the integral of x^n when $n = -1$ we have $\int x^{-1} dx = \frac{1}{x}$

since $\frac{d}{dx} \ln x = \frac{1}{x}$. We then conclude that $\int \frac{1}{x} dx = \ln |x| + c$



Examples

Compute the following integrals

$$1. \int_{-6}^{-2} \frac{1}{x} dx$$

$$2. \int_1^5 \frac{6}{x} dx$$

$$3. \int \frac{1}{x+2} dx$$

$$4. \int \frac{4x}{x^2+5} dx$$

$$5. \int \frac{1}{2x+3} dx$$

$$6. \int \frac{2x}{3x^2-2} dx$$

$$7. \int \frac{x^3}{x^4+10} dx$$



Example

Compute the following integrals

$$1. \int \frac{5}{(x-2)(x+3)} dx$$

$$2. \int \frac{4}{x^2-4} dx$$

$$3. \int \frac{2x-1}{(x+1)^2} dx$$

$$4. \int \frac{x}{4-x^2} dx$$

$$5. \int_2^3 \frac{5+x}{(1-x)(5+x^2)} dx$$

$$6. \int_2^3 \frac{x-4}{(x+2)(x-1)} dx$$



Example

Evaluate the following integrals, correct to 3 *sf*

$$1. \int_3^5 \frac{2}{x^2-1} dx$$

$$2. \int_{-1}^0 \frac{2}{(x-1)(1+x^2)} dx$$

$$3. \int_2^3 \frac{x-9}{x(x-1)(x+3)} dx$$



Example

Find the following integrals

$$1. \int \frac{x}{1+x^2} dx$$

$$2. \int \frac{x}{1+x} dx$$

$$3. \int \frac{x}{1-x^2} dx$$

$$4. \int \frac{x-2}{x^2-4x-5} dx$$

$$5. \int \frac{7x+2}{3x^3+x^2} dx$$

$$6. \int \frac{x}{16-x^2} dx$$



Integration of Exponential Functions

We recall that $\frac{d}{dx}(e^x) = e^x$. Thus $\int e^x dx = e^x + c$.

In general, if $f(x)$ is a differentiable function of x which yields a constant upon differentiation then

$$\int e^{f(x)} dx = \frac{1}{f'(x)} \cdot e^{f(x)} + c.$$

Also $\int e^{kx} dx = \frac{1}{k} e^{kx} + c.$



Examples

1. $\int e^{2x} dx$

2. $\int (2 - e^{2x}) dx$

3. $\int 10xe^{x^2} dx$

4. $\int (2x + 1)e^{2x^2+2x} dx$

5. $\int e^{2x+3} dx$

6. $\int x^2 e^{x^3} dx$



Application of Integration

1. **Average Value of a Function:** We recall that the average/mean of n numbers $x_1, x_2, x_3, \dots, x_n$ is given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1, x_2, x_3, \dots, x_n}{n}.$$

Now the average value of a continuous function f , on the closed interval $[a, b]$ is given by

$$\frac{1}{b-a} \int_a^b f(x) dx .$$



Example

Determine the average value of $f(x) = x^3 - 2x^2 - 5x + 1$ on the interval $[-2, 4]$

Example

The annual per capita consumption of alcoholic beverages on KNUST campus can be modeled by the function

$$f(x) = 43e^{-0.012x}, \quad 1 \leq x \leq 11,$$

where x is measured in years. Determine the average per capita consumption of alcoholic beverages on campus.



2. **Area under a Curve:** Suppose $y = f(x)$ is the equation of a curve and that it is required to find the area enclosed by the curve, the x-axis, and the lines $x = a$ and $x = b$. This is given by

$$A = \int_a^b f(x) dx .$$

The area is positive if it is above the x-axis, and negative if it is below the x-axis



Examples

1. Determine the area enclosed by the curve $y = 2x + 3$, the x-axis and the ordinates $x = 1$ and $x = 4$.
2. Find the area enclosed by the curve $y = 3x + 4$, the x-axis and the lines $x = 1$ and $x = 4$.
3. Determine the area bounded by the curve $y = x^2 + 5$, the x-axis and the lines $x = 0$ and $x = 3$.
4. Sketch $y = x^2 - 4x - 5$ and find the area bounded by the curve and the x-axis.
5. Determine the area enclosed between the curve $y = x(x - 1)(x - 2)$ and the x-axis.



3. The Area between two Curves: The area enclosed between two curves $y = f_1(x)$ and $y = f_2(x)$ is given by

$$A = \int_a^b f_2(x) dx - \int_a^b f_1(x) dx$$

Examples

1. Sketch the curves $y = x^2 + 1$ and $y = 7 - x$ on the same axis and find the area enclosed between the curves.
2. Determine the coordinate of the points of intersection of the curves $y = x^2$ and $y^2 = 8x$. hence sketch the curves $y = x^2$ and $y^2 = 8x$ on the same axis and calculate the area enclosed by the two curves.
3. On the same axis, sketch and label clearly the curves $y = x^2 - 5$ and $y = 5 - x^2$. Find the area of the finite region enclosed by the two curves.



4. Finding the equation of a curve when the gradient function is given:

When the gradient function $\frac{dy}{dx}$ of a curve $y = f(x)$ is given, we can find the equation of the curve by integration. Thus if

$$\frac{dy}{dx} = h(x)$$

then $dy = h(x) dx$. Therefore $\int dy = \int h(x) dx$ and
 $y = \int h(x) dx + c$.



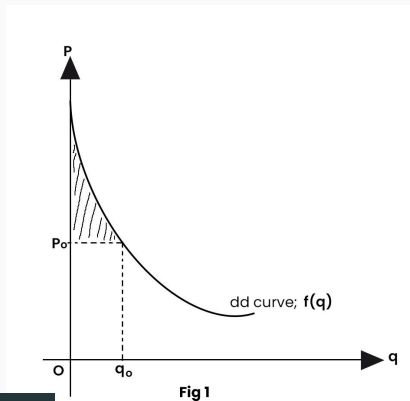
Examples

1. A curve passes through the point $(1, 0)$. If the gradient function at any point $P(x, y)$ is $3x^2 - 1$, find the equation of the curve.
2. The gradient function of a curve at any point $P(x, y)$ is $4x^2 - 3x + 1$. If the curve passes through the point $A(1, 0)$, find the equation of the curve.
3. The gradient function of a curve at any point is given by $(2x - 3x^2)$. If the curve passes through the point $(1, 2)$, find the equation of the curve.
4. A curve passes through the point $(1, -11)$ and its gradient function at any point is $ax^2 + b$, where a and b are constants. A tangent to the curve at the point $(2, -16)$ is parallel to the x-axis. Find;
 - a. the values of a and b ,
 - b. the equation of the curve.



Consumers' and Producers' Surplus

1. **Consumers Surplus:** It is the benefit enjoyed by consumers who are willing and able to offer prices above the equilibrium price. This benefit is shown as the shaded area in the diagram below.



Let p_0 and q_0 be the equilibrium price and the quantity demanded respectively. If the demand function is $p = f(q)$, then consumers surplus (CS), is given by

$$C.S = \int_0^{q_0} f(q) dq - p_0 q_0$$



Examples

1. The demand function for a product is $p = 100 - 0.5q$, where p is the price and q is the quantity demanded. If the supply function is $p = 10 + \frac{q}{10}$, determine the consumers surplus
2. In each of the cases below, determine the total benefit/gain to consumers who are willing to pay more than the equilibrium price.

a. $p_d = 20 - 0.8q$

$$p_s = 4 + 1.2q$$

b. $p_d = 900 - q^2$

$$p_s = 100 + q^2$$

c. $p_d = \frac{50}{q+5}$

$$p_s = \frac{q}{10} + 4.5$$

d. $p_d = 400 - q^2$

$$p_s = 200q + 100$$

e. $q = 100(10 - p)$

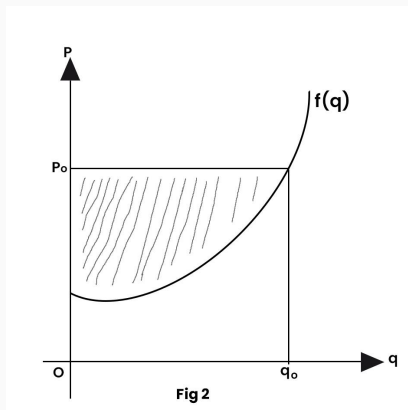
$$q = 80(p - 1)$$

f. $q = \sqrt{100 - p}$

$$q = \frac{p}{2} - 10$$



2. **Producers Surplus:** There is an economic gain to producers who are willing to supply their products at prices lower than the equilibrium price p_0 . The total gain to producers is the shaded area shown in the diagram below



Let p_0 and q_0 be the equilibrium price and the quantity supplied respectively. The shaded area depicts the total gain to producers. If the supply function is $p = f(q)$, then producers surplus (PS), is given by

$$P.S = p_0 q_0 - \int_0^{q_0} f(q) dq$$

Example

The demand function for a product q is $p = \frac{90}{q+2}$ and the supply equation is $p = q + 1$. Determine the producer's surplus.



Examples

1. Given the demand function $p = 42 - 5q - q^2$ and assuming that the equilibrium price is GH¢6. Find the consumers' surplus.
2. Find the consumers' surplus when equilibrium price and quantity are respectively GH¢32.5 and 25 and $p = 45 - 0.5q$
3. Given that $p = (q + 3)^2$, find the producers' surplus at $p_0 = 81$ and $q_0 = 6$.
4. Given the demand function $p = 25 - q^2$ and the supply function $p = 2q + 1$, assuming that there is pure competition in the economy, find
 - (a) the consumers' surplus,
 - (b) producers' surplus.



Exercise

1. Given the demand function $p = 113 - 3q^2$ and the supply function $p = (q + 1)^2$ under pure competition, find the consumers' and the producers' surplus.
2. Under monopoly, the quantity sold and market prices are determined by the demand function. If the demand function for a profit maximizing monopolist is $P = 274 - q^2$ and $MC = 4 + 3q$. find the consumers' surplus.



Remark

For profit maximization, $MC = MR$

Proof

Let π denote the profit function. Then $\pi = R - C$, where R = total revenue and C = total cost

Now from $\pi = R - C$,

$$\frac{d\pi}{dq} = \frac{d}{dq}(R) - \frac{d}{dq}(C)$$

But for maximum profit $\frac{d\pi}{dq} = 0$

$$\therefore \frac{d}{dq}(R) - \frac{d}{dq}(C) = 0$$

$$\implies \frac{dR}{dq} = \frac{dC}{dq}$$

But $\frac{dR}{dq} = MR$ and $\frac{dC}{dq} = MC$

$\therefore MR = MC$ as required.



Differential Equations

A differential equation is an equation that contains an unknown function and one or more of its derivatives. The order of a differential equation is the order of the highest derivative that occurs in the equation.

Examples are $\frac{dy}{dx} = xy$, $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 4x^2$, $\frac{d^2y}{dx^2} = -ax$

The general solution of a differential equation is a function that satisfies the differential equation together with a constant of integration. This is called the general solution. If conditions that enable us to determine the value of the constant of integration are given then the solution is called a particular solution. The conditions are called initial condition



Separable Differential Equations

A differential equation of the form

$$\frac{dy}{dx} = f(x) \cdot g(y)$$

or

$$\frac{dy}{dx} = \frac{f(x)}{g(y)}, \quad g(y) \neq 0$$

is separable if everything involving x can be moved to one side of the equation and everything involving y can be moved to other side of the equation. For example $\frac{dy}{dx} = \frac{x^a}{y^b}$ is a first-order separable differential equation. The method of solving such a differential equation is called separation of variables.



Example

Find the general solution of the following differential equations

1. $\frac{dy}{dx} = 3x^2$

2. $\frac{dy}{dx} = \frac{x^2}{y^3}$

3. $\frac{dy}{dx} = 2x$

4. $\frac{dy}{dx} = xy$

5. $x^2 \frac{dy}{dx} = y(y - 1)$

6. $\frac{dy}{dx} = \frac{x}{y}$

7. $\frac{dy}{dx} = \frac{y}{x}$

Example

Determine the particular solution for the following differential equations

1. $\frac{dy}{dx} = 3xy; \quad x = 0, \quad y = 3$

2. $\frac{dy}{dx} = xy - 3x; \quad \text{at } y(0) = 1$



Example

The annual rate of increase in the number of employees between 1950 and 1985 at the NewMedia Company can be modeled by the differential equation

$$\frac{dy}{dt} = 0.3t^{0.6}, \quad 0 \leq t \leq 35,$$

where t represents the number of years and y represents the number of employees in 100s

1. If the company had 1800 employees in 1965, find the employee growth function for the company
2. Use your answer in (1) to estimate the number of employees at the company in 1983.



Example

The StrikeNow Company determines that the marginal cost for their MaxiPak of lighters is given by the differential equation

$$\frac{dC}{dx} = -0.02x + 6, \quad 0 \leq x \leq 300,$$

where x is the number of MaxiPaks produced and C is the cost in dollars.

1. Determine the general solution of the cost function C
2. Find the particular cost function, if the cost of producing 10 Paks is 400.



Exercise

Determine the general solution of the following differential equations.

1. $\frac{dy}{dx} = -2$

2. $\frac{dy}{dx} = e$

3. $\frac{dy}{dx} = 1 - 3x$

4. $\frac{dy}{dx} = \sqrt{x} + 2$

5. $\frac{dy}{dx} = \frac{5}{x^2}$

6. $\frac{dy}{dx} = 10x^4 - x^2 + 3$



Exercise

Determine the particular solution of the following differential equations.

1. $\frac{dy}{dx} = 12x; \quad x = 1, y = 8$
2. $\frac{dy}{dx} = 3x^2 + 4x; \quad y(1) = 6$
3. $\frac{dy}{dx} = xe^{x^2}; \quad y(0) = 2$
4. $\frac{dy}{dx} = x^3 - 2x; \quad y(0) = -2$



Partial Differentiation

So far we have taken a look at functions of one independent variable and their related rates. If $y = f(x)$ then $\frac{dy}{dx}$ is the total derivative of y with respect to x . Now if $z = f(x, y)$ then the derivative of z with respect to either x or y is called the partial derivative of z . An example of the form $z = f(x, y)$ represents a function of two independent variables. The domain of $z = f(x, y)$ is the set of all ordered pairs (x, y) for which z is defined.



Examples

1. Consider $f(x, y) = 2x^2 - 3y^3$. Determine the following
 - a. $f(2, 1)$
 - b. $f(-1, 2)$
 - c. $f(1, -2)$
 - d. $f(2, 3)$
2. The cost of producing x tennis balls and y volleyballs is given by $C(x, y) = 0.97x + 0.89y$
 - a. Determine the domain of $C(x, y)$.
 - b. Evaluate $C(250, 700)$ and interpret it.



Definition of Partial Derivatives

1. Let $f(x, y)$ be a function of two variables. The partial derivative of $f(x, y)$ with respect to x is defined as

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}.$$

The partial derivative of $f(x, y)$ with respect to x is found by treating y as a constant and performing our ordinary differentiation techniques with respect to x . The notation for this partial derivative is f_x or $\frac{\partial f}{\partial x}$.

2. The partial derivative of $f(x, y)$ with respect to y is defined as the limit

$$\frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$



This is obtained by treating x as a constant and performing our ordinary differentiation techniques with respect to y . The notation for this partial derivative is f_y or $\frac{\partial f}{\partial y}$.



Examples

1. Determine $f_x(x, y)$ and $f_y(x, y)$ for $f(x, y) = x^y - y^2$.
2. Compute the partial derivatives for $f(x, y) = x^2y + y^3x - 2xy = y$.
3. Determine f_x and f_y for each of the following.
 - a. $f(x, y) = 5x^2 - 6y^3$
 - b. $f(x, y) = 2xy - y^2 + 1$
 - c. $f(x, y) = x^3y^2$
 - d. $f(x, y) = x^2y^4$
 - a. $f(x, y) = \frac{x}{y}$



Applications of the Partial Derivatives

1. Analysis of Marginal Revenue

Example; A company sells two products: military boots and life jackets. Let x and y be the number of military boots sold per month and the number of life jackets sold per month respectively. Suppose that $p_1 = 1200 - 18x - y$ and $p_2 = 523 - 1.5x - 0.15y$ are the demand function for the military boots and life jackets respectively. Determine

1. the revenue function $R(x, y)$.
2. $R_x(x, y)$ and $R_y(x, y)$ and interpret it,
3. evaluate and interpret $R_x(50, 25)$.



Example

1. Donyma Steel Complex spends x thousand cedis each week on newspaper advertising and y thousand cedis each week on radio advertising. The company has weekly sales, in terms of thousands of cedis, given by $S(x, y) = 2x^2 + y$. Determine
 - a. $S_x(x, y)$ and $S_y(x, y)$.
 - b. $S_x(3, 5)$ and $S_y(3, 5)$ and interpret each.
2. If $S(x, y) = 3x + 2y^3$, compute $S_x(2, 6)$ and $S_y(2, 6)$ and interpret each.
3. For $f(x, y) = x^2 + y^2 - 8x + 2y + 7$, determine the value of x and y such that $f_x = 0$ and $f_y = 0$ simultaneously.



2. Marginal Productivity of Labour and Capital

If L represents the units of labour and K represents the units of capital, then the total production Q is given by the Coff-Douglas production function.

$$Q(L, K) = AL^a K^b,$$

where A , a and b are constants.

Example

If $f(x, y) = 4x^{0.75}y^{0.25}$, determine the realistic domain of $f(x, y)$ and evaluate $f(x, y)$ when $x = 50$ and $y = 60$ and interpret it.



Definition

For any function $Q(x, y) = Ax^a y^b$ where A , a and b are constants and x and y represent units of labour and capital respectively, Q_x gives the approximate change in productivity per unit change in labour. It is called the marginal productivity of labour, and Q_y gives the approximate change in the production per unit change in capital. Q_y is called the marginal productivity of capital



Example

The production function for a company is given by $f(x, y) = 1.64x^{0.64}y^{0.4}Q$, where x is the number of labour hours in hundreds and y is capital in thousands, and Q is the quantity in 1000s of gallons.

1. Compute f_x and f_y .
2. Find $f_x(100, 18)$ and interpret it.
3. Find $f_y(110, 20)$ and interpret it.



Remark

For some amounts invested in labour and in capital, we wish to know whether production would increase more rapidly if additional resources were invested in labour or in capital. We can approximate the change in productivity by any of the following

1. The change in productivity with respect to labour = $f_x \Delta x$
2. The change in productivity with respect to capital = $f_y \Delta y$

Example

Suppose that a production function is given by

$$f(x, y) = 4.23x^{0.37}y^{0.66},$$



where x =labour (million) and y =capital (million). Currently, $x = 5$ and $y = 1$. Would production increase more by spending additional GH¢1m on labour or GH¢500000 on capital equipment?

